

# Ali Shafiee

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[alishafiee.github.io](https://alishafiee.github.io) | [Codeforces](#) | [LinkedIn](#) | [IOI Profile](#) | [Google Scholar](#)

*IOI silver medalist and ICPC World Finalist. MSc researcher in stochastic control, robust optimization, and algorithmic game theory.*

## EDUCATION

### Institute of Science and Technology Austria (ISTA)

MSc in Computer Science, grade average 1.0 (Austrian scale, 1.0 highest)

Vienna, Austria

Sep. 2025 – Sep. 2027 (expected)

### Sharif University of Technology

BSc in Computer Engineering, GPA 3.68/4 (17.03/20)

Tehran, Iran

Sep. 2018 – Aug. 2023

## PUBLICATIONS

Author lists are alphabetical, per convention in theoretical computer science.

- **Randomise Alone, Reach as a Team.** L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026*. [\[paper\]](#)
- **Qualitative Analysis of  $\omega$ -Regular Objectives on Robust MDPs.** A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026*. [\[paper\]](#)
- **Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete.** A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025*. [\[paper\]](#)
- **Quantitative Analysis of  $\omega$ -Regular Robust MDPs.** A. Asadi, K. Chatterjee, E. K. Goharshady, M. Karrabi, A. Montaseri, **A. Shafiee**. *AAAI 2027*, under review. [\[paper\]](#)
- **On the Complexity of Discounted Robust MDPs with  $L_p$  Uncertainty Sets.** A. Asadi, K. Chatterjee, A. Montaseri, **A. Shafiee**. *NeurIPS 2026*, under review. [\[paper\]](#)
- **Mechanism Design for Multi-Resource Nakamoto Consensus.** N. Babashah, M. A. Baig, K. Pietrzak, **A. Shafiee**. *WINE 2026*, under review.

## COMPETITIVE PROGRAMMING

- **Codeforces: top 50 worldwide in 2019**; International Grandmaster, peak rating 2800 (handle ToTLeS)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

## RESEARCH EXPERIENCE

### Institute of Science and Technology Austria (ISTA)

Research Intern and Online Collaborator, then MSc Researcher, advisor *Prof. Krishnendu Chatterjee*

Vienna, Austria

Apr. 2024 – present

- Proved NP-completeness of limit-sure reachability for small-memory POMDP policies, resolving an open complexity question; published at UAI 2025.
- Derived algorithms and complexity bounds for robust MDPs under  $\omega$ -regular and discounted objectives, leading to three further papers (AAAI 2026, NeurIPS 2026, AAAI 2027).
- Implemented solvers in C++ and Python and benchmarked them against existing approaches on random and structured MDP families, scaling to instances with thousands of states.

## INDUSTRY EXPERIENCE

### Quera, Iran's largest competitive-programming and technical-education platform

Contest Manager, Content Manager, Director of CS Education, Business Line Manager

Tehran, Iran

Mar. 2019 – Aug. 2025

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, Django, Linux/Git, data analysis, machine learning, and deep learning.
- Ran 20+ large-scale programming contests, including sponsored hiring hackathons with corporate partners.

## TEACHING & SERVICE

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, **IOI 2019**, Baku
- **Problem-setter** (2017 – present) for the Iranian National Olympiad in Informatics, Codecup and Quera; **olympiad coach** (2017 – 2023), 50+ students trained for INOI and the IOI
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming

## SKILLS

**Programming:** C++, Python, Rust · **Numerical & ML:** NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch, JAX

**Tools:** Git, Linux/Bash, SQL, Docker, Jupyter, L<sup>A</sup>T<sub>E</sub>X · **Mathematics:** probability, stochastic processes, Markov decision processes, optimization, game theory

**Languages:** Persian (native), English (TOEFL 110/120)